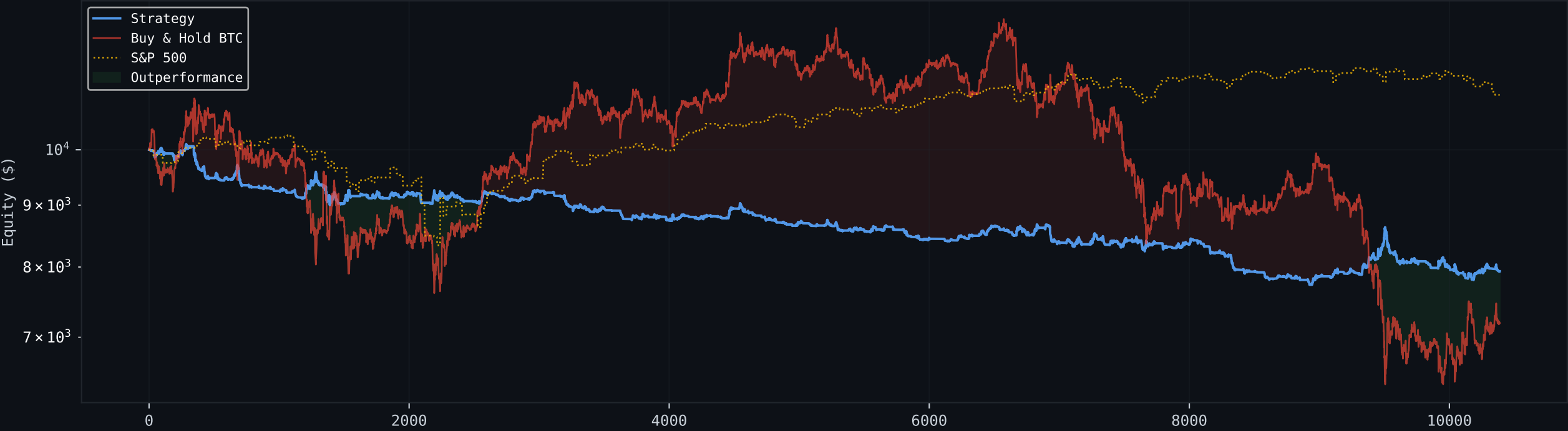


INSTITUTIONAL QUANTITATIVE TEARSHEET
CNN+LSTM PPO × SPA (Pure OOS Walk-Forward)

Sharpe: -1.62 | Sortino: -1.68 | Calmar: -0.75 | Max DD: -23.5% | Alpha p-value: 0.9595 | Beta: -0.056

Equity Curve (Log Scale) vs. Benchmark & S&P 500



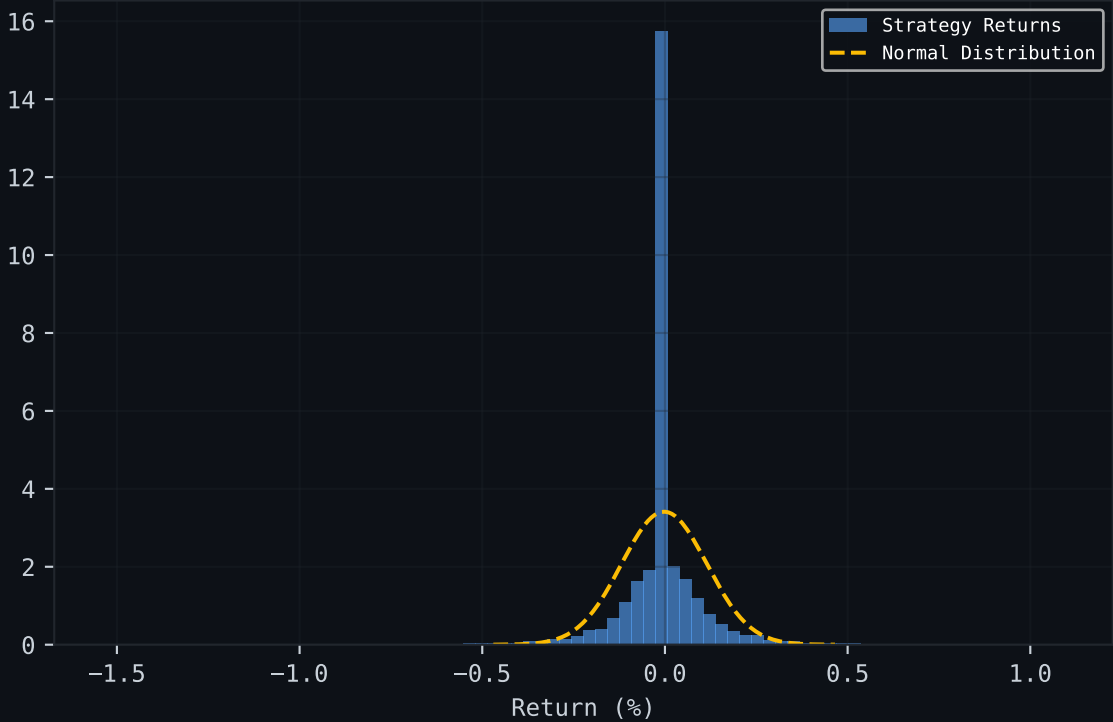
Underwater Plot (Drawdown %)



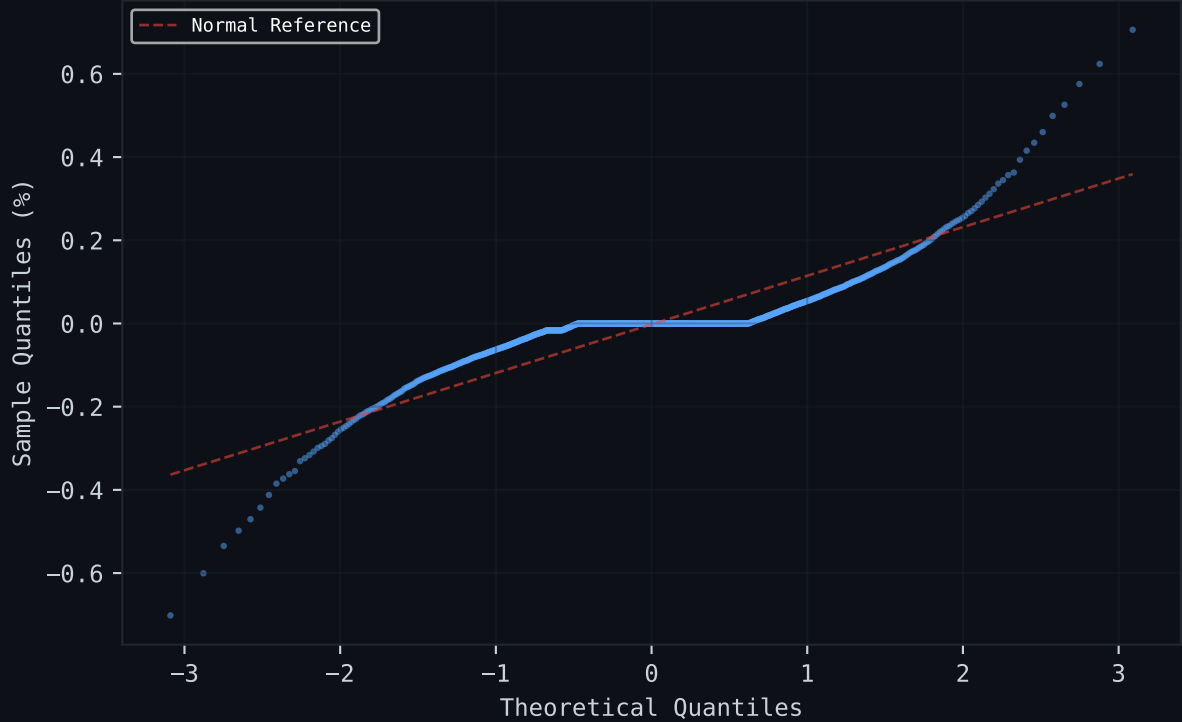
Rolling Sharpe Ratio (window=2596 bars $\approx$ 108d)



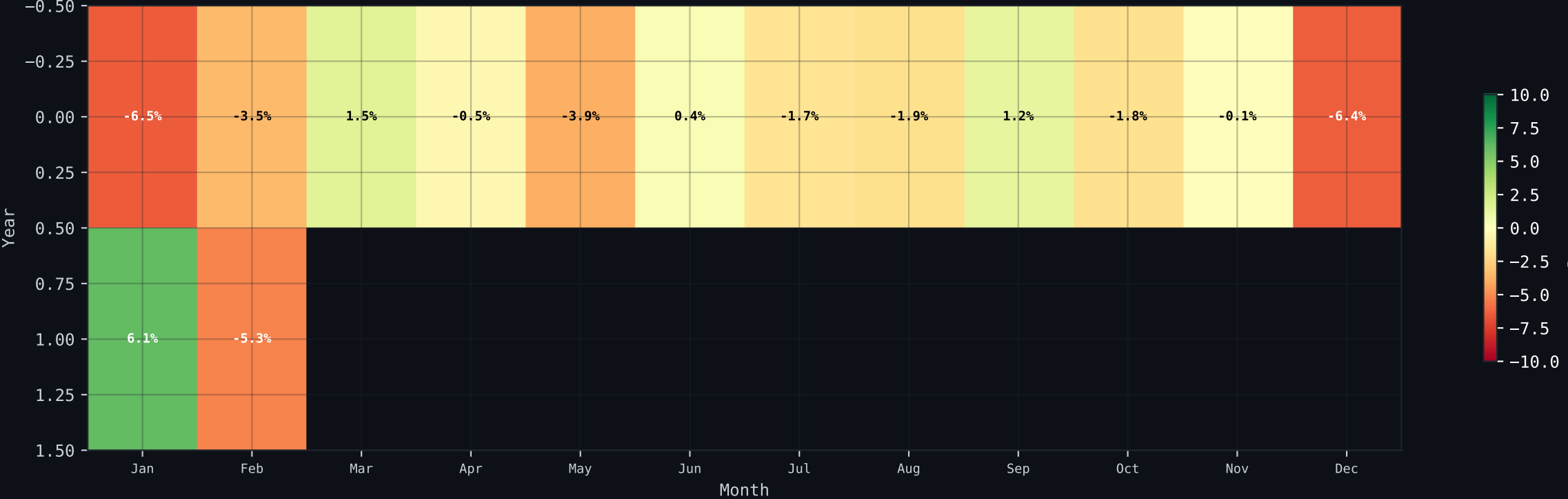
Return Distribution vs. Normal



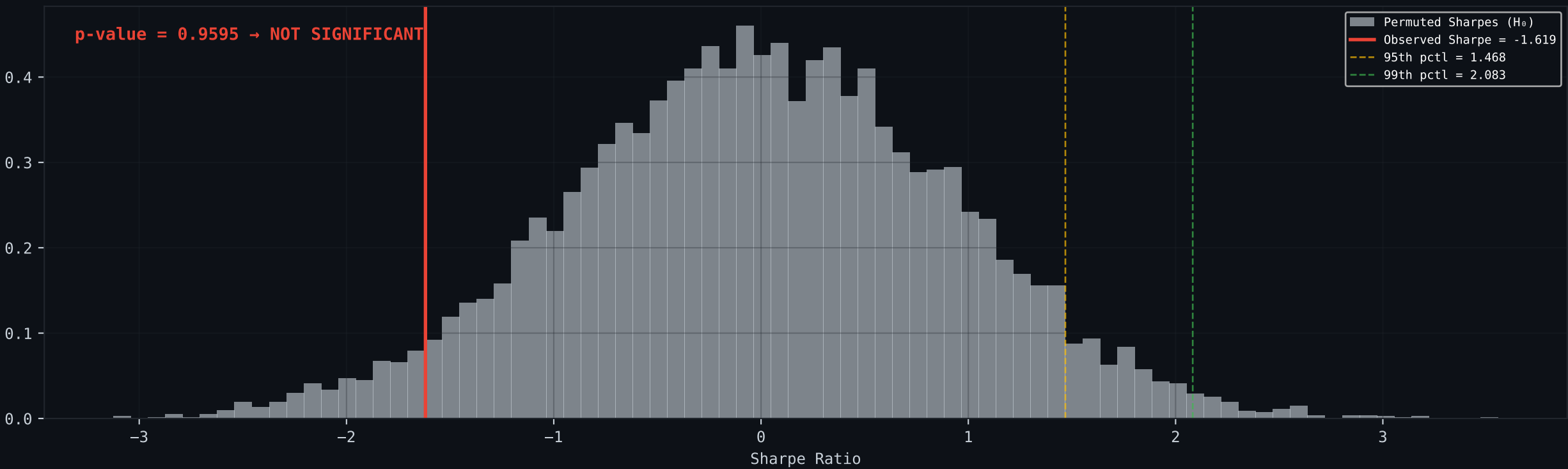
Q-Q Plot (Fat Tail Detection)



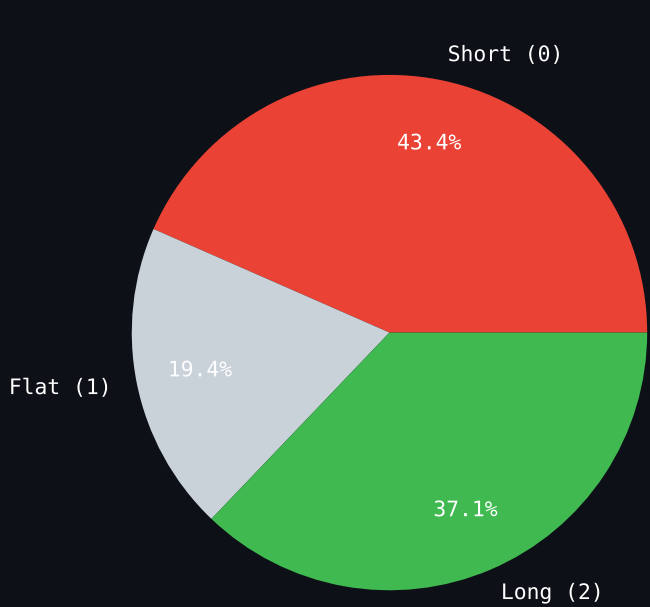
Monthly Returns Heatmap (%)



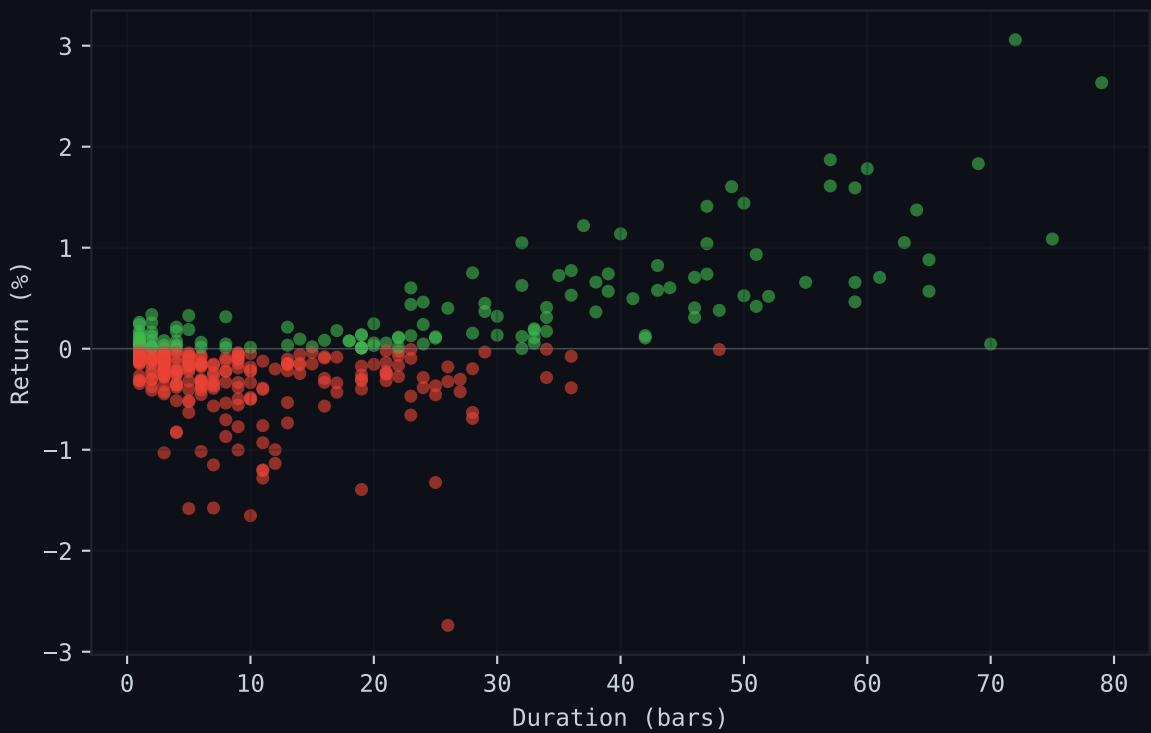
Monte Carlo Permutation Test (10,000 simulations)



Action Distribution



Trade Returns vs. Duration



QUANTITATIVE RISK METRICS – FULL DETAIL

Metric	Value
PERFORMANCE METRICS	
Total Return	-20.64%
Annualized Return	-17.71%
Annualized Volatility	10.94%
Sharpe Ratio	-1.619
Sortino Ratio	-1.680
Calmar Ratio	-0.754
Omega Ratio (θ=0)	0.929
Profit Factor	0.929
Win Rate	45.7%
BENCHMARK COMPARISON	
BTC Buy & Hold Sharpe	-0.527
BTC Buy & Hold Sortino	-0.671
BTC Buy & Hold Max DD	-50.10%
S&P 500 Sharpe	0.517
S&P 500 Sortino	0.112
S&P 500 Max DD	-18.90%
TAIL RISK METRICS	
Max Drawdown	-23.50%
VaR (95%)	-0.1714%
VaR (99%)	-0.3623%
CVaR / Expected Shortfall (95%)	-0.2955%
CVaR / Expected Shortfall (99%)	-0.5255%
Skewness	-0.150
Excess Kurtosis	17.000
ALPHA DECOMPOSITION	
Alpha (per period)	-0.000023
Alpha t-statistic	-2.034
Alpha p-value	0.0420
Beta (benchmark exposure)	-0.0560
R² (benchmark correlation)	0.0558
STATISTICAL VALIDATION	
Permutation Test p-value	0.9595
Significance (p < 0.05)	NO ☐
CAPACITY ESTIMATION	
Est. Capacity (USD)	\$0
Current Market Impact	0.28 bps
Participation Rate	0.000008%